Breast Cancer Diagnosis

Predicting malignancy from cell nucleus measurements with supervised and unsupervised learning

MU5BM753 Supervised and Unsupervised learning for biology

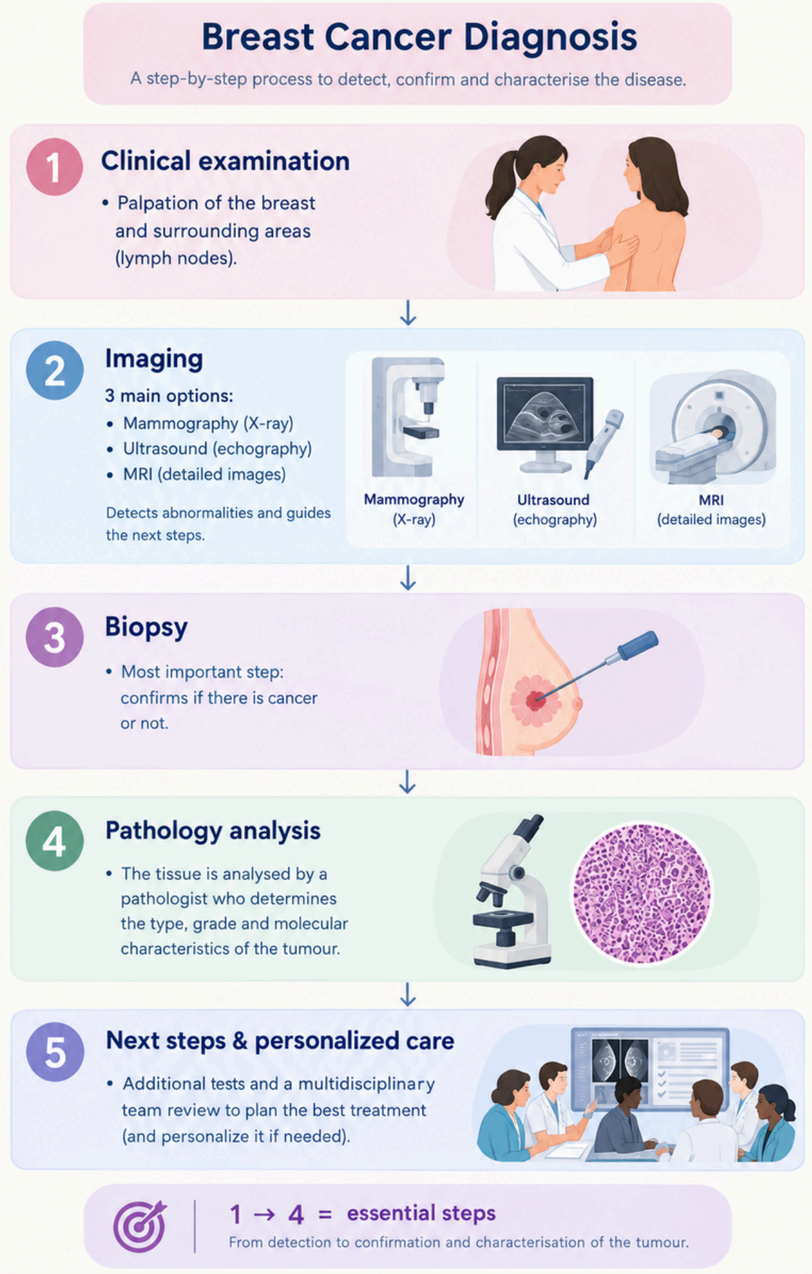
Ann-Solène Adelson, Anaïs Hluszko and Corentin Osta

Breast cancer : overview

- **Mammary epithelial cells** acquire alterations that **disrupt normal** control of proliferation, survival, genome maintenance and tissue organization
- These changes also modify the **cell's phenotype**, in particular, **nuclear appearance**
- Malignant cells typically show:
  - Enlarged, irregular nuclei
  - Greater variation in nuclear size and shape
  - Disrupted chromatin patterns

➔ **cytological criteria used by pathologist**[1,2]

- **Diagnosis**: made by a pathologist reading the slide by eye
- **Limitation**: human assessment can vary between pathologists[3] and takes a long time
- **Goal**: use nuclear features for a fast, quantitative assessment



One hypothesis, tested two ways

➔ **Nuclear morphology alone carries enough signal to reveal and predict malignancy**

Supervised approach

Can a classifier reliably :

- **Help the pathologist** by providing a **rapid preliminary assessment** of the likelihood of malignancy ?
- **Support the diagnostic** assessment ?
- **identify** which nuclear characteristics drive that prediction the most ?

Can quantified nuclear atypia (size, contour, texture, ...) offer the pathologist a more reproducible second opinion ?

Unsupervised approach

Are the **nuclear characteristics** alone sufficient to separate most of the samples into **distinct groups**, particularly benign and malignant tumors ?

Is malignant transformation a strong enough signal on its own to create two distinct, separable cell populations ?

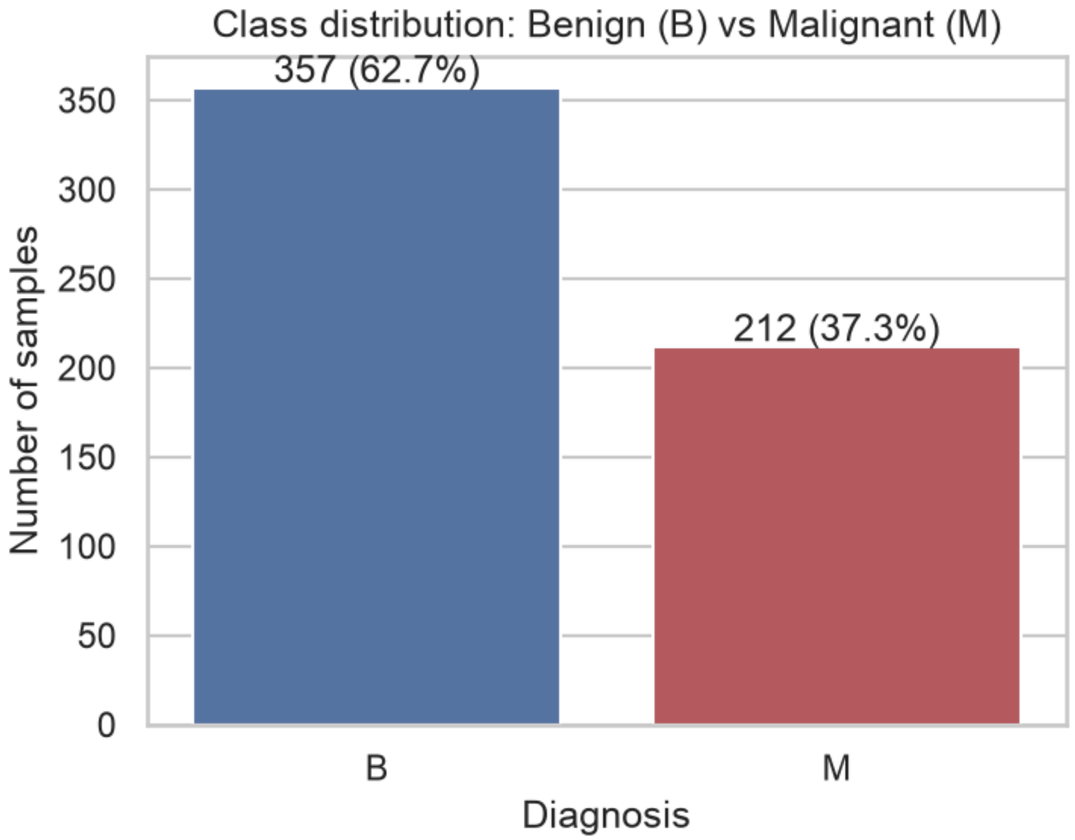
The Wisconsin Breast Cancer dataset

569
tumor samples

30
measured features

- Each sample: **10 base measurements** of a cell nucleus (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension),
- Each is summarized by 3 statistics: mean, standard error, and worst (the 3 most extreme cells).
- The classes are imbalanced: 63% benign, 37% malignant.

Quality checks confirm the data is clean: no missing values, no duplicate samples.

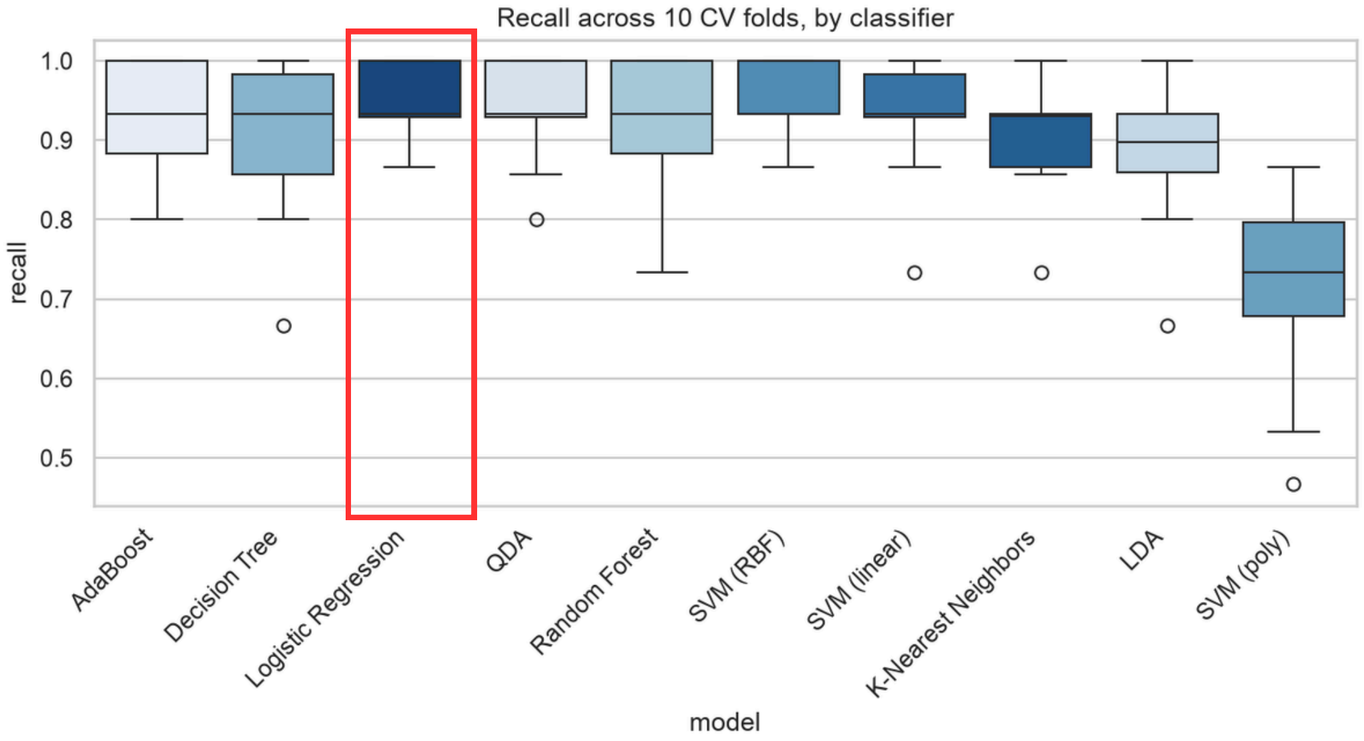


Bioinformatic workflow

1. **Quality checks**
2. **Exploratory Analysis**
3. **Initial Screening**
elects models by **average rank** and confirmed by **recall/ARI**
4. **Feature selection**
Different methods used
5. **Supervised approach**
6. **Unsupervised approach**

Average rank : ranks each model per metric, then averages
Recall : proportion of malignant cases caught
ARI : agreement with the true diagnosis, corrected for chance

Logistic Regression is the strongest supervised baseline



	model	accuracy_mean	precision_mean	recall_mean	f1_mean	f2_mean	roc_auc_mean	avg_rank_across_6_metrics
0	Logistic Regression	0.977308	0.986607	0.952381	0.968440	0.958569	0.992610	2.00
1	SVM (RBF)	0.974872	0.980000	0.952857	0.965475	0.957691	0.992667	2.67
2	AdaBoost	0.967372	0.980833	0.932381	0.954261	0.940643	0.994305	3.67

- **10 classifiers** are compared with 10-fold **cross-validation** on 6 metrics

~0.95

Mean Recall for the **Logistic Regression**

2.00

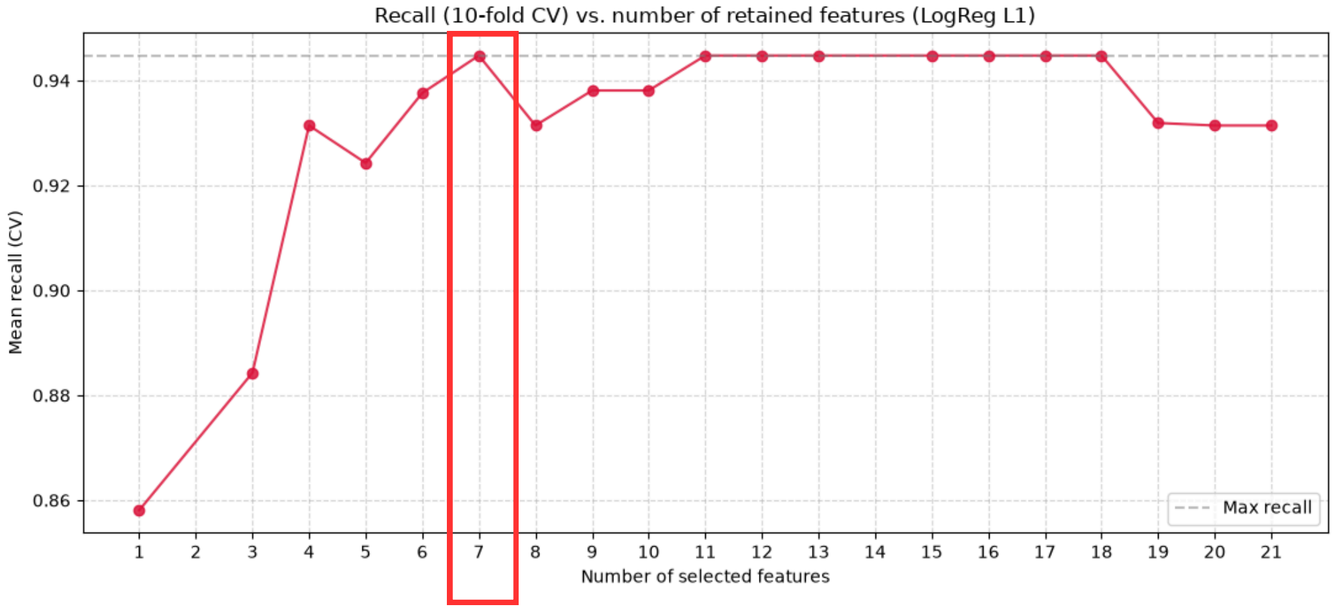
Average rank across 6 metrics for **Logistic Regression**

This **screening** does **two things** at once:

- **Picks** which classifier to deploy
- **Fixes** which model will drive some of the embedded feature selection

7 of 30 features are enough to predict malignancy

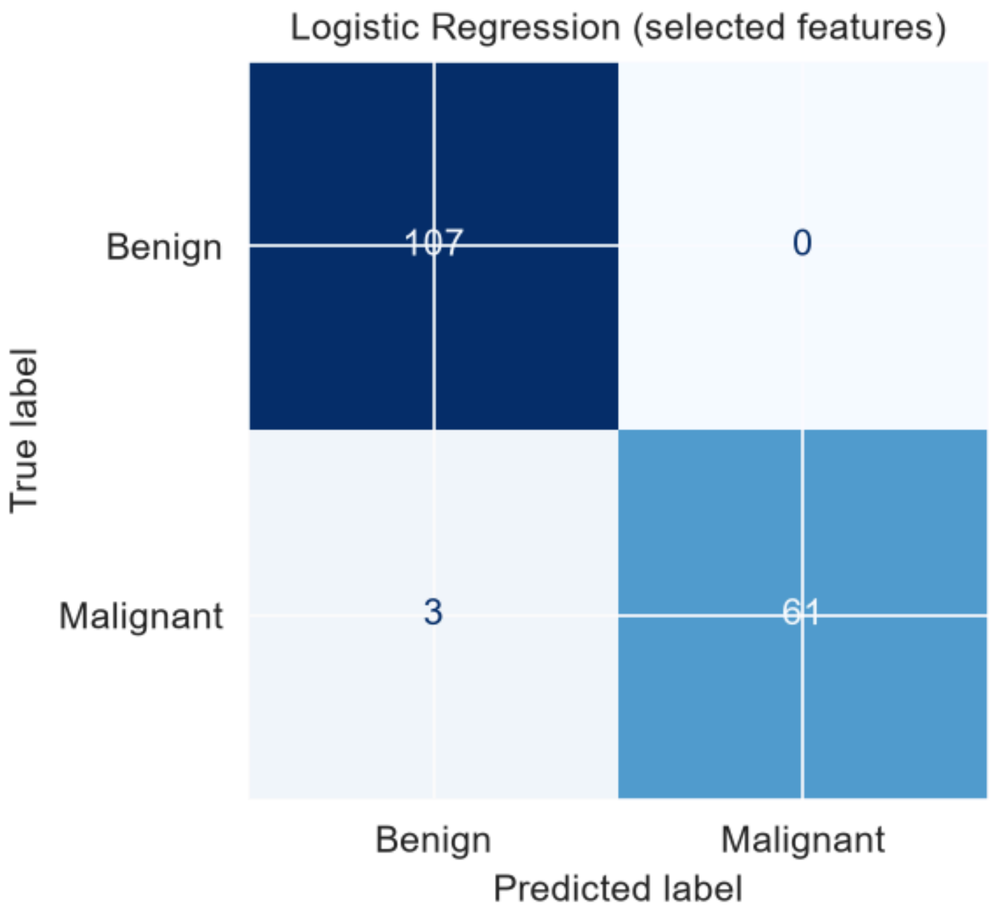
Method	Family	Features	Recall	Avg. rank
LogReg L1	embedded	7	94.5%	2.0
SelectFdr	filter	18	93.8%	2.4
LinearSVC L1	embedded	9	93.8%	2.9
Elastic-Net	embedded	6	93.8%	3.8
Variance Threshold	filter	23	93.1%	4.0



- **7 collinear features** (correlation above 0.95) are **removed** first.
- 5 **selection methods** are then benchmarked, from two families:
  - **Filter** : scores each feature on its own.
  - **Embedded** : selects features while training a model.
- **LogReg L1** wins : best **average rank** (2.0) and strongest **recall** (94.5%).

The 7 selected features :
radius_mean, concave_points_mean, radius_se, texture_worst, smoothness_worst, concave_points_worst, symmetry_worst

Tuned model misses only 3 malignant cases out of 64



- Tuned by grid search using the 1-SE rule (C = 0.3857) on the 7 selected features.
- Evaluated on the held-out test set, never seen during training or tuning.

95.3%

Recall: only 3 malignant cases missed out of 64

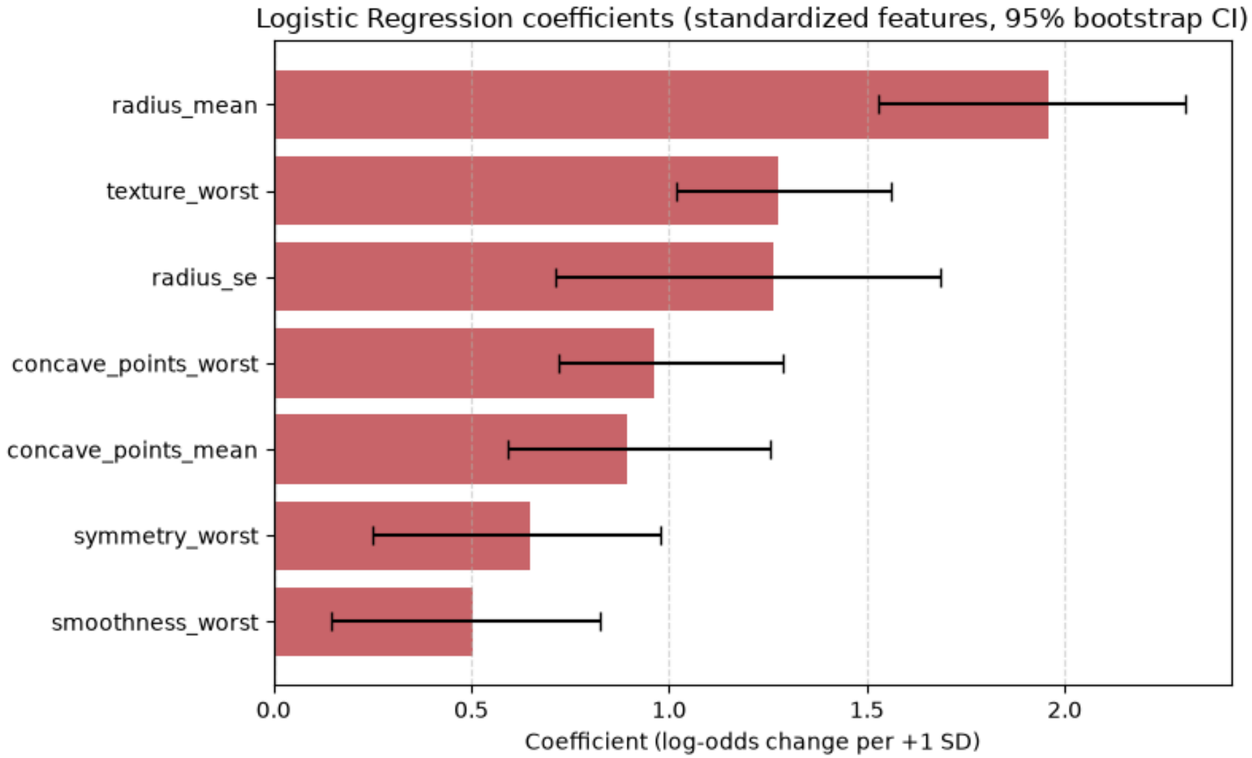
100%

Precision: no false alarm among 61 flagged cases

A **classifier** reliably predicts malignancy from a **selected** set of **measurements**.

Nuclear Size and Texture Drive the Model's Decision

	coefficient	ci_low	ci_high	odds_ratio_per_SD
radius_mean	1.960	1.528	2.306	7.096
texture_worst	1.274	1.018	1.563	3.576
radius_se	1.263	0.711	1.687	3.537
concave_points_worst	0.964	0.720	1.288	2.622
concave_points_mean	0.895	0.593	1.257	2.448
symmetry_worst	0.649	0.248	0.977	1.914
smoothness_worst	0.504	0.145	0.827	1.656

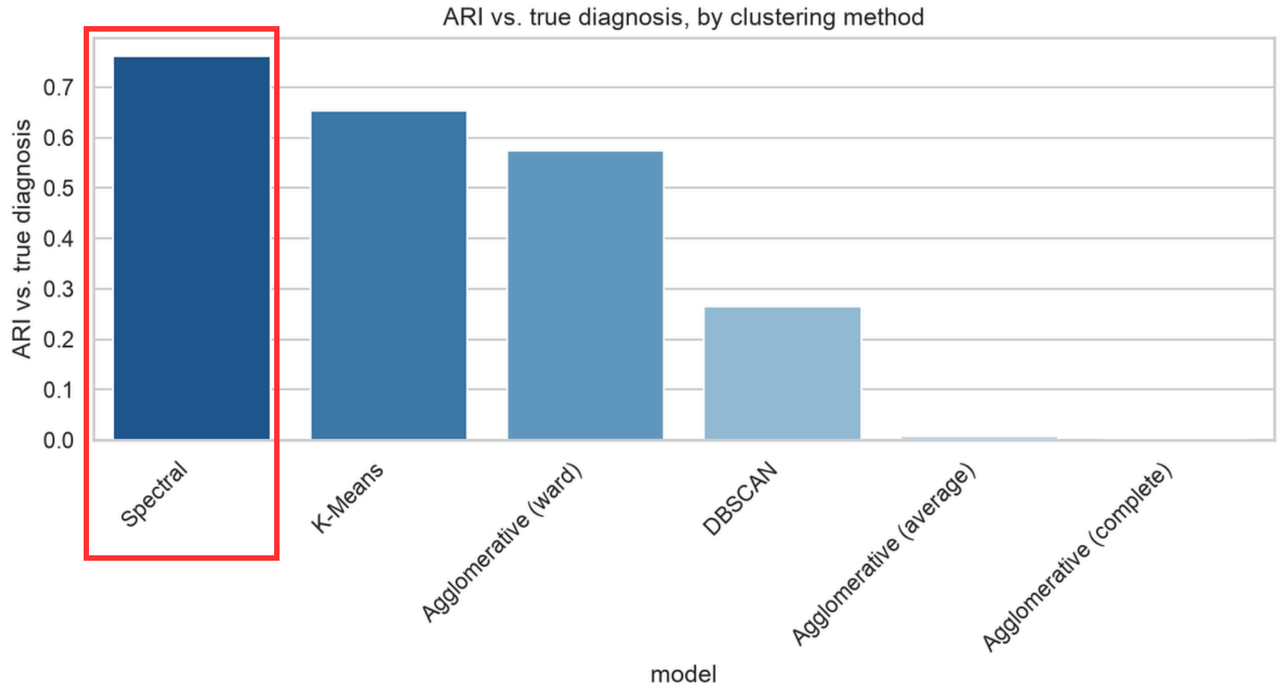


- Log-odds coefficients evaluated on standardized data.
- 95% Confidence Intervals validated via 1,000 bootstrap iterations.

Dominant driver : *radius_mean*
Increases the odds of malignancy by a factor of 7.1 per additional standard deviation.

Reliable features
No confidence interval crosses zero. Every selected feature actively contributes to the final prediction.

Spectral Clustering recovers best the true diagnosis



model	silhouette	homogeneity	completeness	v_measure	ARI_vs_diagnosis	avg_rank_across_5_metrics
Spectral	0.336653	0.649723	0.676197	0.662696	0.760801	1.8
K-Means	0.343382	0.524888	0.540147	0.532408	0.653625	2.2
omerative (ward)	0.339385	0.446214	0.468142	0.456915	0.575041	3.2

- **6 clustering methods** group the samples without ever using the diagnosis
- Their output is then **checked** against the true labels using **ARI**

0.76

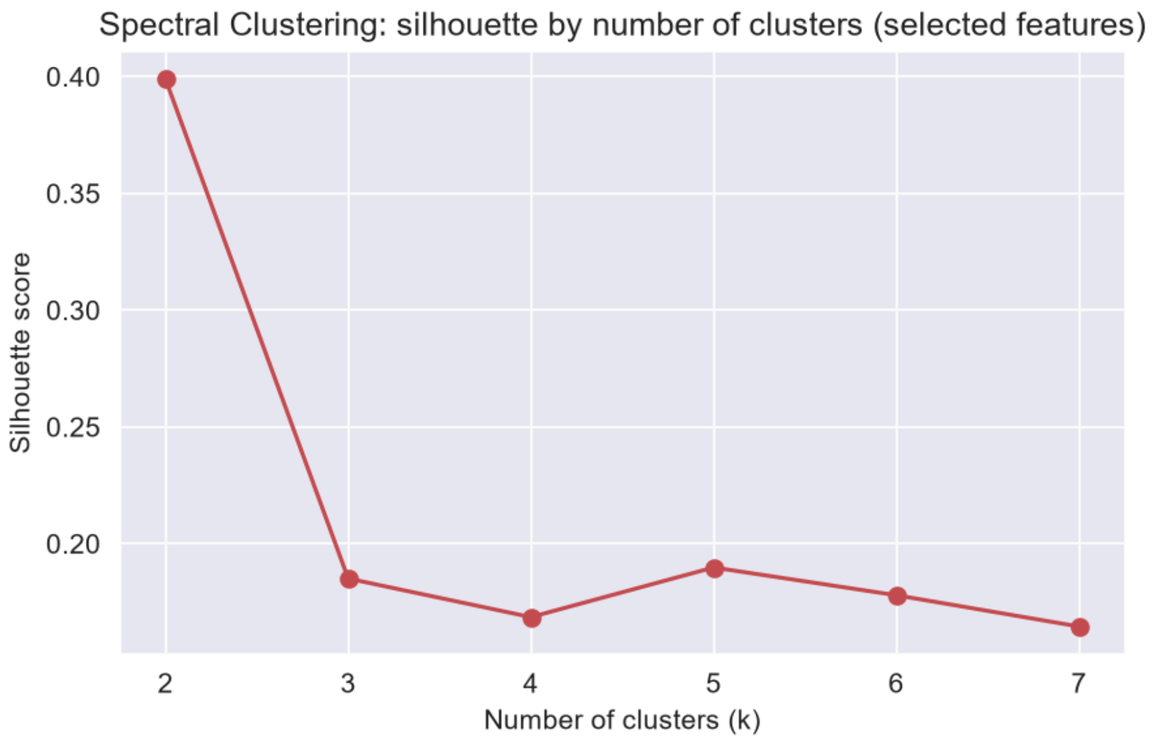
Highest ARI for **Spectral clustering**

1.8

Average rank across 5 metrics for **Spectral clustering**

- This **screening picks** which clustering method to deploy.

Two clusters capture the main structure



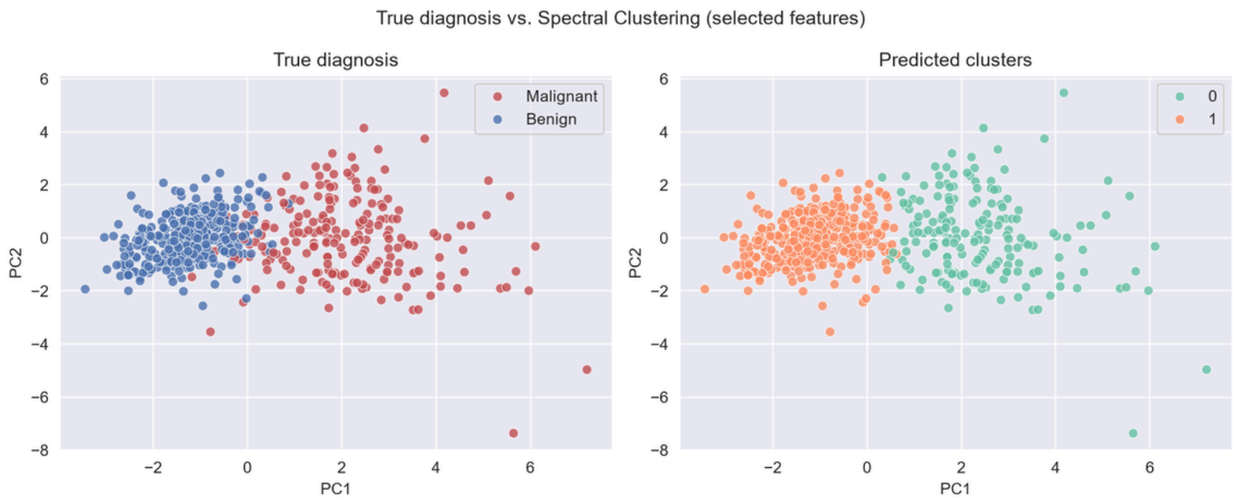
- The **screening** above **assumed** 2 groups, matching the 2 known diagnosis classes.
- Here, k is swept from 2 to 7 using **silhouette**

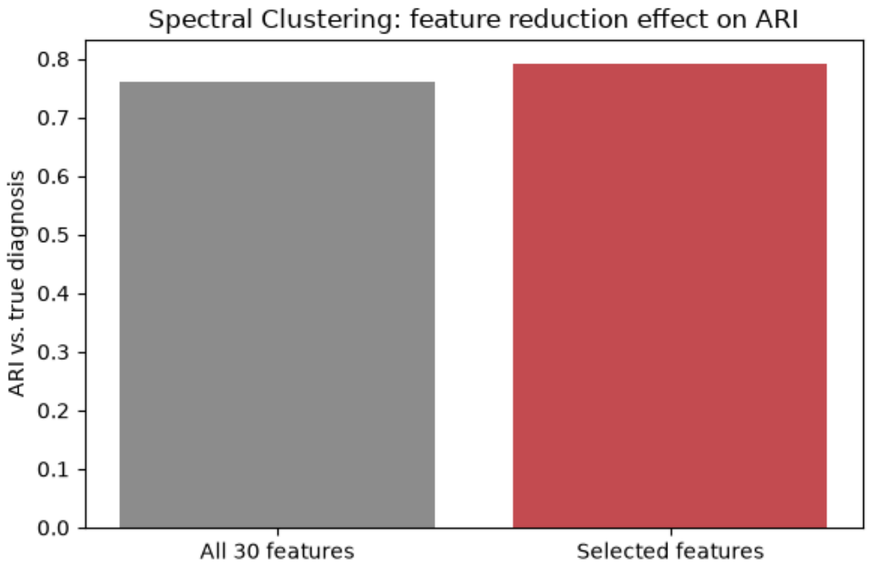
Silhouette peaks sharply at k = 2

0.4 at k = 2, dropping to 0.16–0.19 for every k from 3 to 7. More clusters reveal no additional structure.

- The **data** itself **confirms** the **assumption**, without ever seeing the labels:
  - **malignant** and **benign nuclei** each form **one coherent group**,
  - with **no sign of hidden sub-types** within either class.

Feature selection strengthens the clustering–diagnosis match





- The **PCA projection** shows the predicted clusters closely matching the true diagnosis.
- Restricting to the **7 selected features** raises the ARI from **0.761** to **0.792**

0.792

ARI vs true diagnosis, without the clustering ever seeing the labels

Malignancy alone is a **strong enough signal** to organize the data into 2 groups, without ever using the labels.

The 7 features mirror criteria pathologists already use

Size
(**radius_mean, radius_se**)

- Malignant nuclei tend to be larger and more heterogeneous in size[6]
- Cancer cells can have abnormal chromosome numbers (aneuploidy) which can increase DNA content

Contour irregularity
(**concave_points_mean, concave_points_worst, smoothness_worst**)

- These features capture how irregular the nuclear boundary is
- Irregular nuclear outlines and pleomorphism are important criteria for assessing malignancy[7]

Symmetry & texture
(**symmetry_worst, texture_worst**)

- Abnormalities in the shape and distribution of pixel intensities within the nucleus
- Differences in chromatin organization and nuclear appearance[6]

4 of the 7 features are "worst" statistics: malignancy shows up most clearly in a sample's most abnormal-looking cells, not its average shape.

Strong results still come with limitations

Same Blind Spot as Pathologists

- Some invasive cancers keep regular, benign-looking nuclei — invisible to pure geometry, whether by model or by eye.
- The samples both methods missed (false negatives, misclassified clusters) fit exactly this profile.

➔ a second opinion for clear cases, not the ambiguous ones

Single-Institution Generalizability

Features are derived from a localized, specific imaging setup. The model's exact mathematical thresholds require multi-center validation on modern equipment before any broader clinical use.

External validation is the natural next step

Testing the 7-feature model on the WPBC cohort

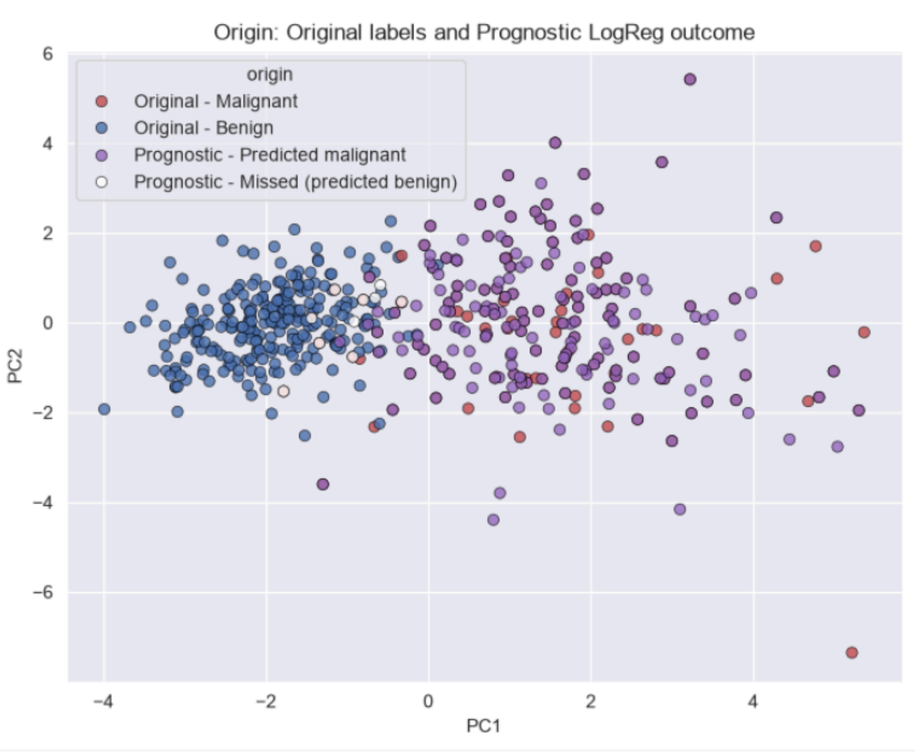
- **Mathematical limitations:** Absence of benign samples precludes the calculation of *precision*, *specificity*, and *false-positive rates*.

- **Analysis of the 13 missed cases:**
  - **Morphological overlap:** False negatives mathematically and visually mimic healthy tissues.
  - **Independent Confirmation:** This benign resemblance is strictly confirmed by both the model's decision weights and unsupervised clustering outputs.

Next Step, Diagnosis to Prognosis: Transitioning from binary diagnosis to predictive tracking by correlating morphological deviations with outcome (recurrence) and time-to-recurrence.

93.4%

Recall: only 13 malignant cases missed out of 198



One hypothesis confirmed two ways

Supervised approach

Logistic Regression, tuned on 7 selected features :

- misses only **3 malignant cases out of 64** (recall 95.3%),
- with **no false alarm** (precision 100%).

Unsupervised approach

Spectral Clustering recovers :

- **2 natural groups** that closely match the diagnosis (ARI 0.792), without ever using the labels
- and **testing more clusters** finds no further structure.

- **Nuclear morphology** alone is enough to both **predict** and **reveal** malignancy
- Reducing it to 7 non-redundant features, out of 30, sharpens both views at once

➔ they **align** with criteria pathologists already use.

However :

- **Both methods** struggle with **borderline cases**, just like **pathologists**
- They can **support diagnosis** in **clear-cut cases**, but do not replace expert judgment in difficult cases

References

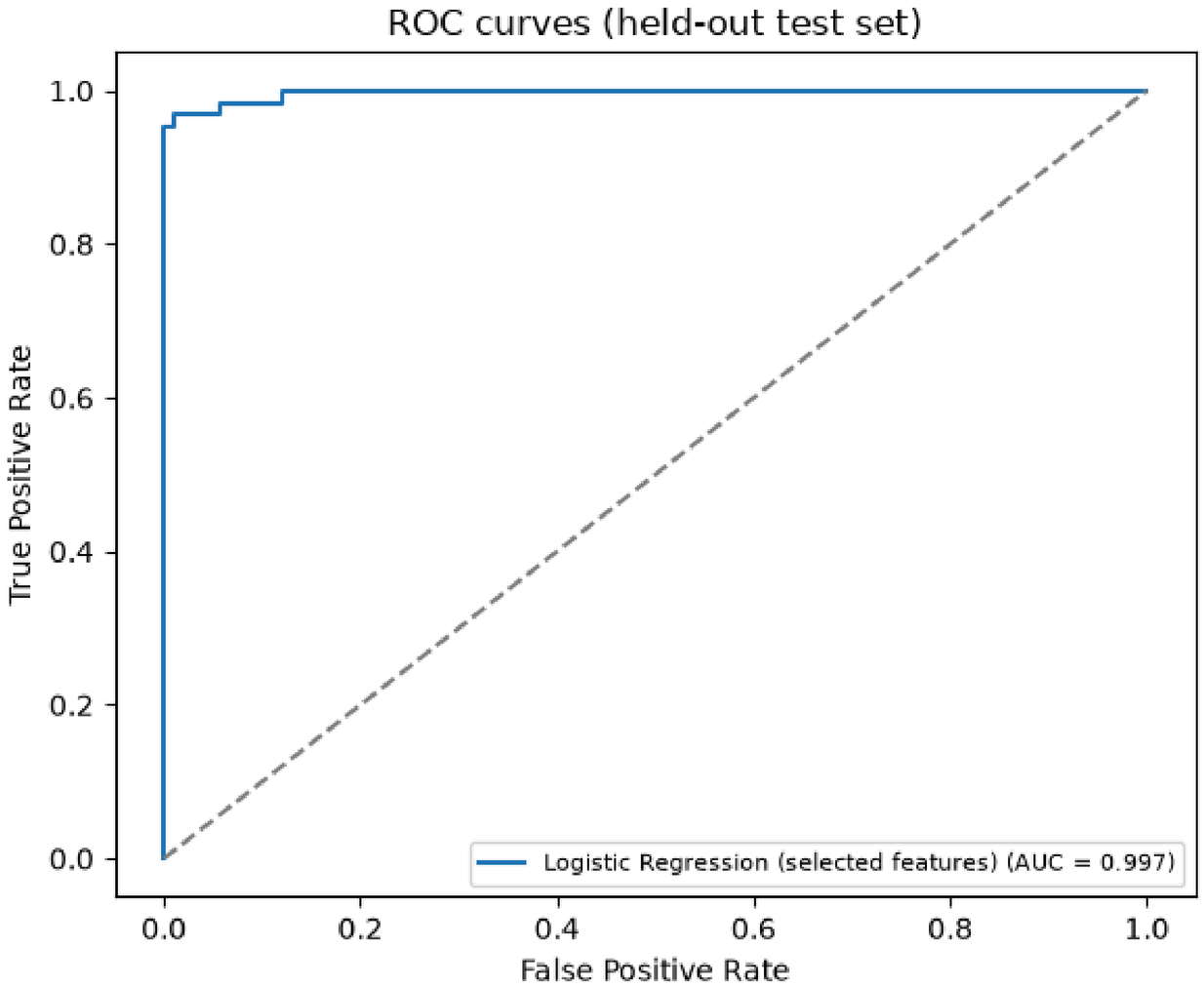
1. National Cancer Institute. How Breast Cancer Is Diagnosed. cancer.gov/types/breast/diagnosis
2. Bussolati, G., Marchiò, C., Gaetano, L., Lupo, R., & Sapino, A. (2008). Pleomorphism of the nuclear envelope in breast cancer: a new approach to an old problem. Journal of Cellular and Molecular Medicine, 12(1), 209–218.
3. Elmore, J. G., Longton, G. M., Carney, P. A., et al. (2015). Diagnostic concordance among pathologists interpreting breast biopsy specimens. JAMA, 313(11), 1122–1132.
4. Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. Proceedings of SPIE, 1905, 861–870.
5. UCI Machine Learning Repository. Breast Cancer Wisconsin (Diagnostic). doi.org/10.24432/C5DW2B
6. Schirmer, E. C., & De Las Heras, J. I. (Eds.). (2014). *Cancer Biology and the Nuclear Envelope: Recent Advances May Elucidate Past Paradoxes*. Advances in Experimental Medicine and Biology, Vol. 773. Springer New York. https://doi.org/10.1007/978-1-4899-8032-8
7.Kashyap, A.; Jain, M.; Shukla, S.; Andley, M. Study of Nuclear Morphometry on Cytology Specimens of Benign and Malignant Breast Lesions: A Study of 122 Cases. J. Cytol. 2017, 34 (1), 10. https://doi.org/10.4103/0970-9371.197591

Appendix

Features

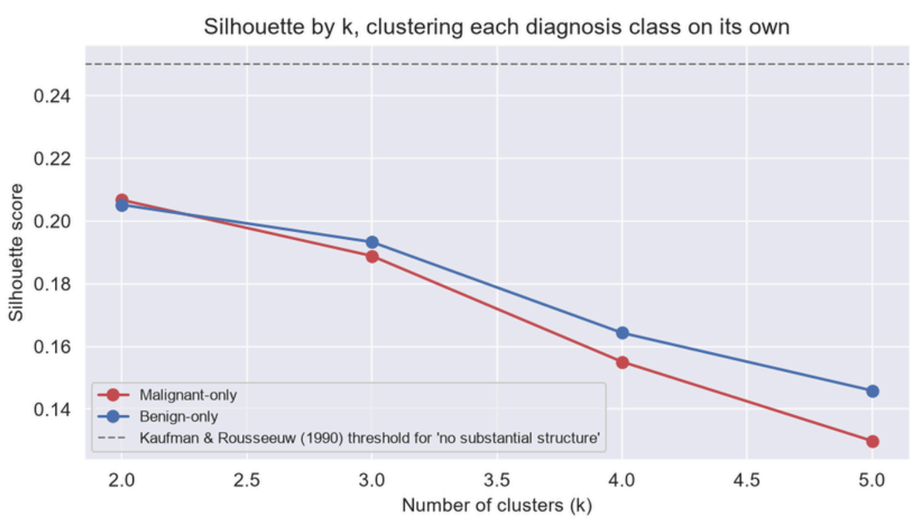
Feature	What it measures
`radius`	Mean distance from the center of the nucleus to its perimeter
`texture`	Standard deviation of gray-scale values. Rougher surfaces give higher values
`perimeter`	Total length of the nucleus's outer contour
`area`	Total 2D surface occupied by the nucleus
`smoothness`	Local variation in radius length. Measures edge regularity
`compactness`	perimeter^2 / area - 1.0. Measures how compact or round the shape is
`concavity`	Severity of the concave, indented portions of the contour
`concave points`	Number of concave portions of the contour
`symmetry`	How symmetric the nucleus shape is
`fractal dimension`	Structural complexity and irregularity of the contour

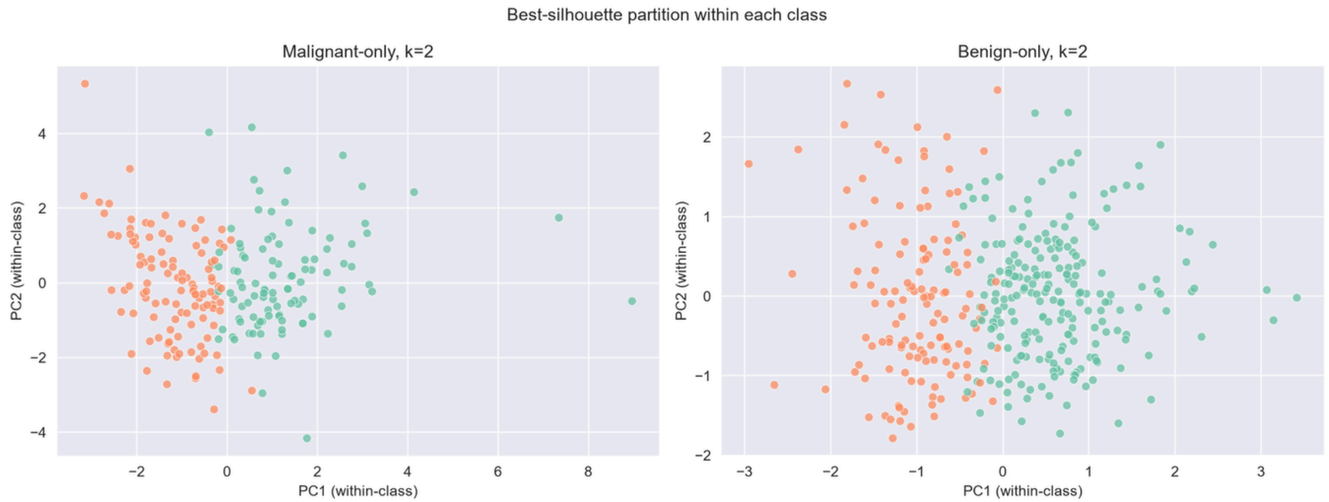
Tuned Logistic Regression Model



ROC-AUC: **0.997**. Near-perfect ranking of malignant above benign cases, across every threshold.

Splitting each class alone confirms it: no hidden sub-type



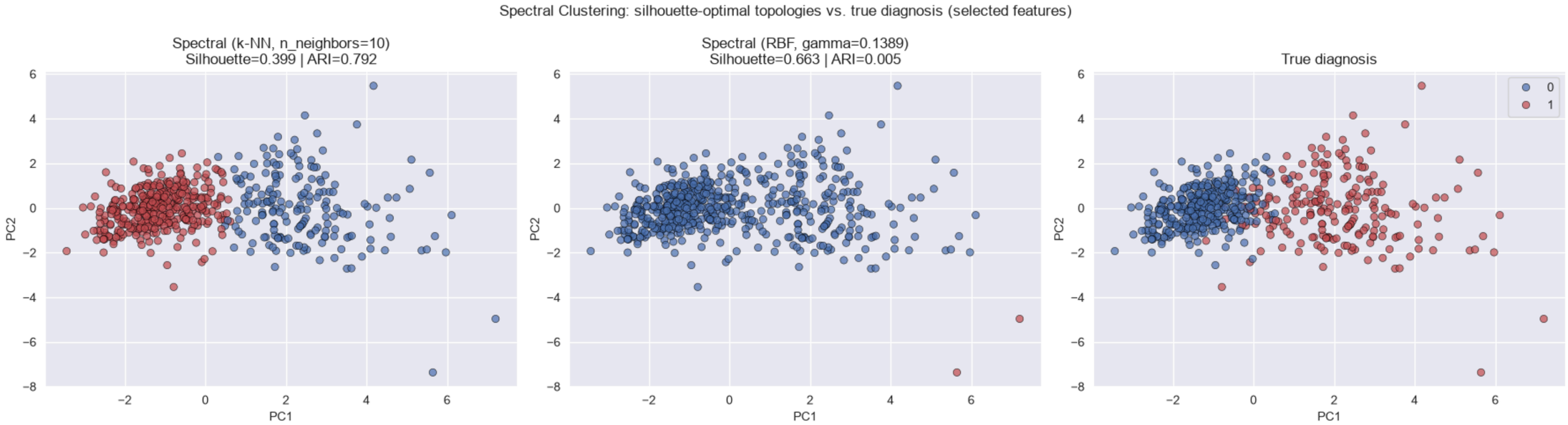


The check above uses the whole dataset. Here, malignant and benign samples are clustered separately, removing the dominant diagnosis signal that could mask a weaker, within-class sub-type.

Both land on a modest split (k = 2)
Silhouette 0.207 (malignant) and 0.205 (benign) ➔ both below 0.25, the threshold Kaufman & Rousseeuw (1990) use for "no substantial structure".

The **PCA plots** confirm this visually: both partitions run along one axis with heavy overlap and no visible gap — a cut through a single continuous cloud, not two distinct sub-groups.

RBF's best silhouette hides a clinically meaningless partition



Both affinity families are swept, then each one's own silhouette-best setting is compared on ARI:

k-NN (n=10)

ARI = 0.792

(silhouette 0.399)

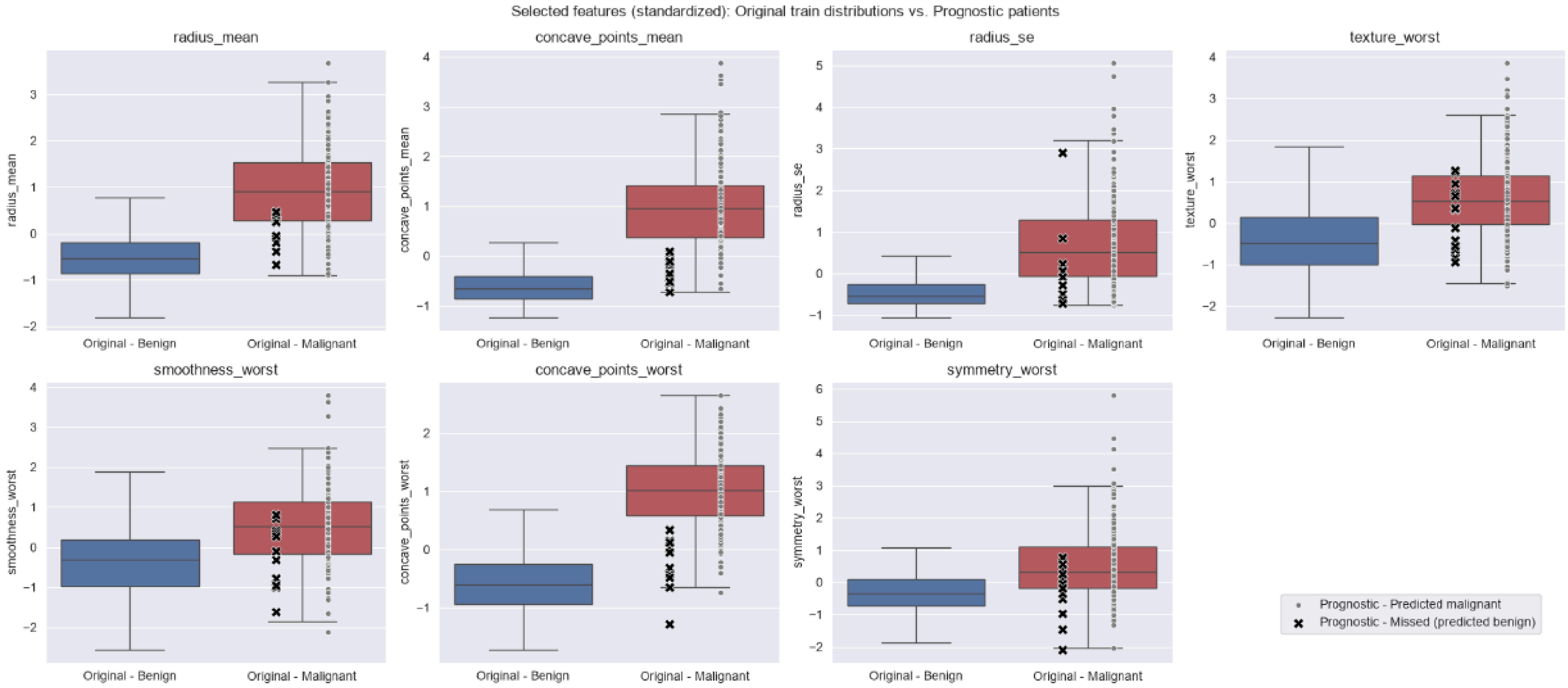
RBF (γ=0.139)

ARI = 0.005

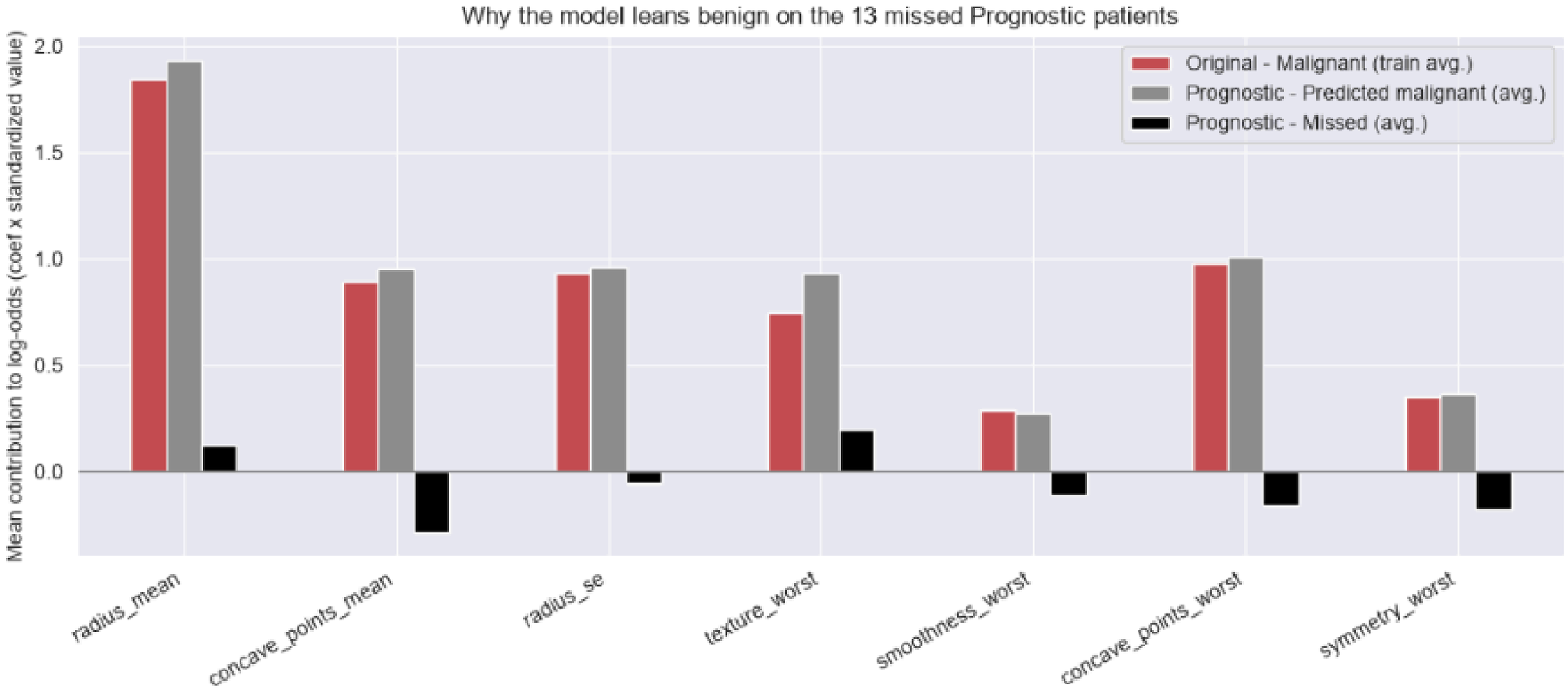
(silhouette 0.663, the highest of the sweep)

RBF's own best silhouette configuration is nearly a random partition against the true diagnosis. k-NN stays stable and clinically aligned across its whole tested range. This is why k-NN, not RBF, is the affinity used throughout.

Understanding False Negatives in External Validation



Understanding False Negatives in External Validation (2)



Implementation details

1. Reusable functions live in `01_scripts/` and are imported by every notebook — no logic is duplicated across the 7 notebooks.

2. The train/test split is made once, in `04_Feature_Selection.ipynb`, and reused unchanged by both "zoom" notebooks — this avoids leaking test information into feature selection or hyperparameter tuning

3. Every random step is fixed (`random_state=42`), so every result in this report is exactly reproducible from the raw data alone.

4. Outputs are separated by type and traceable to the notebook that produced them: tables in `04_results/`, figures in `05_figures/`, filenames prefixed by notebook number.

Difficulties encountered

1 **Logistic Regression: the highest-recall model had unstable coefficients :**
The max-recall `C` (30.4) let correlated features split predictive weight unpredictably between them. Adopting the 1-SE rule (C=0.39) traded a negligible drop in cross-validated recall for coefficients that are stable and interpretable.

2 **External validation needed a scaler that was never saved**
The scaler fit on all 30 features wasn't persisted by `04_Feature_Selection.ipynb`. For the external cohort (WPBC), it had to be reconstructed deterministically — same seed, same split — rather than refit on new data, which would have placed WPBC in a different feature space than the model was trained in.